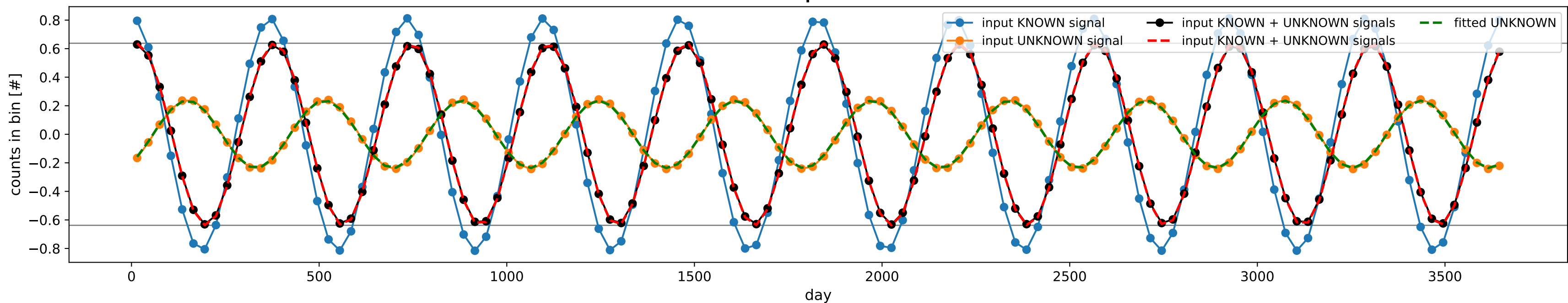
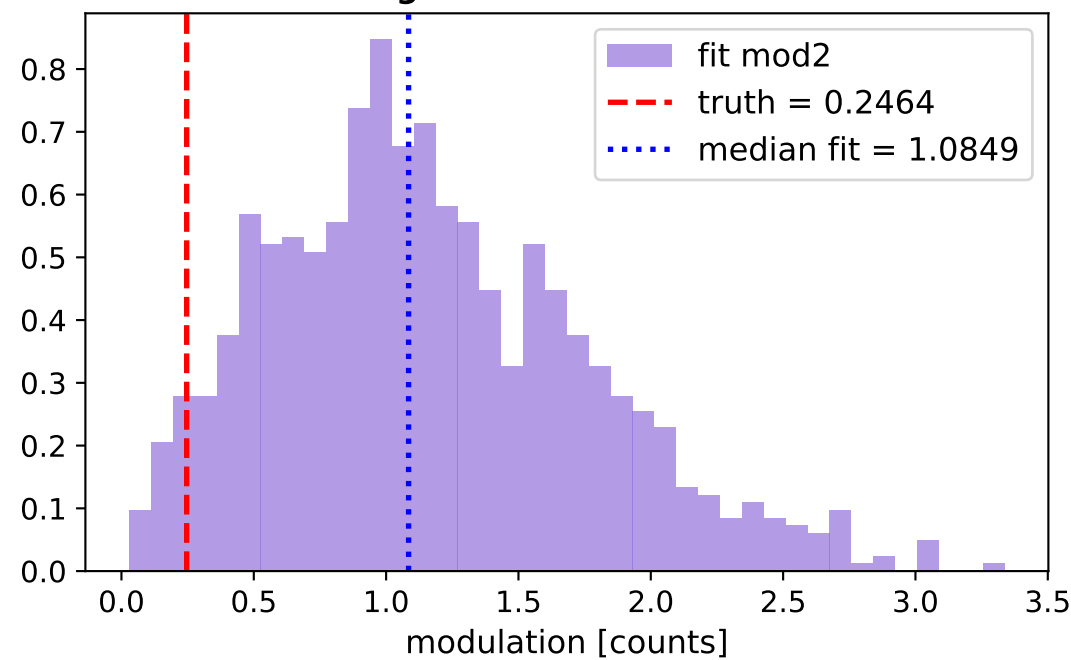
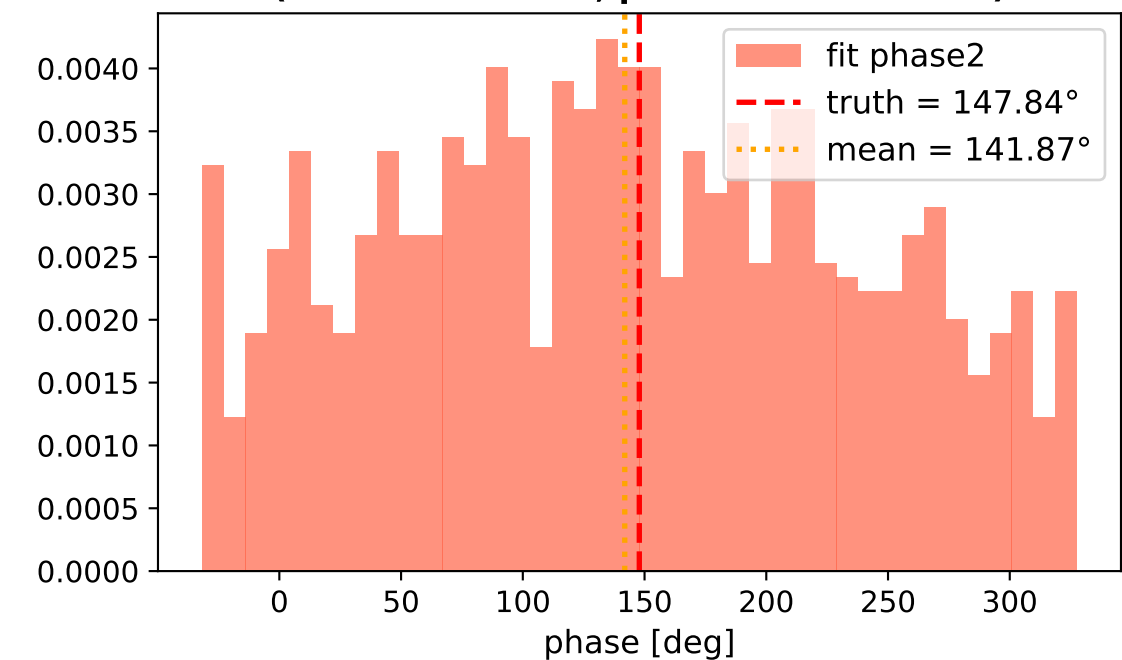


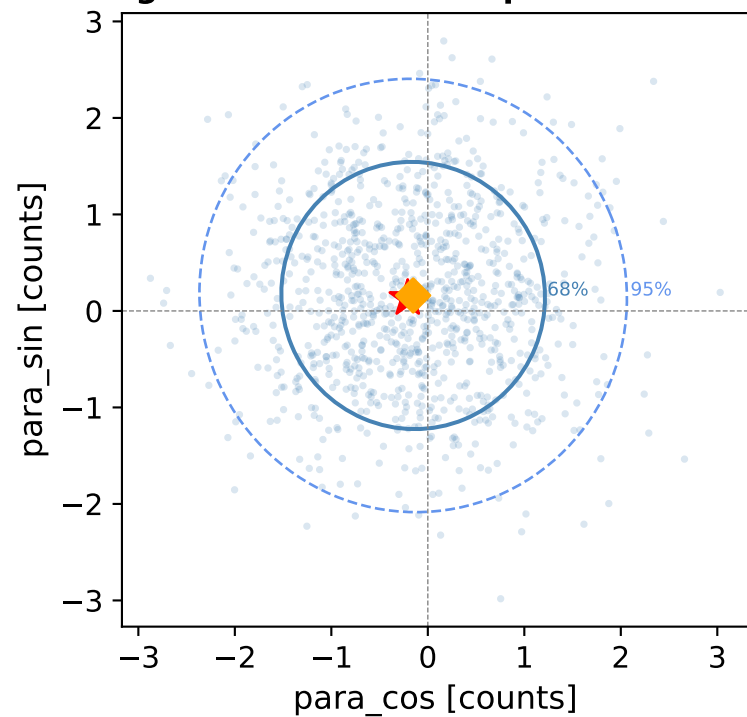
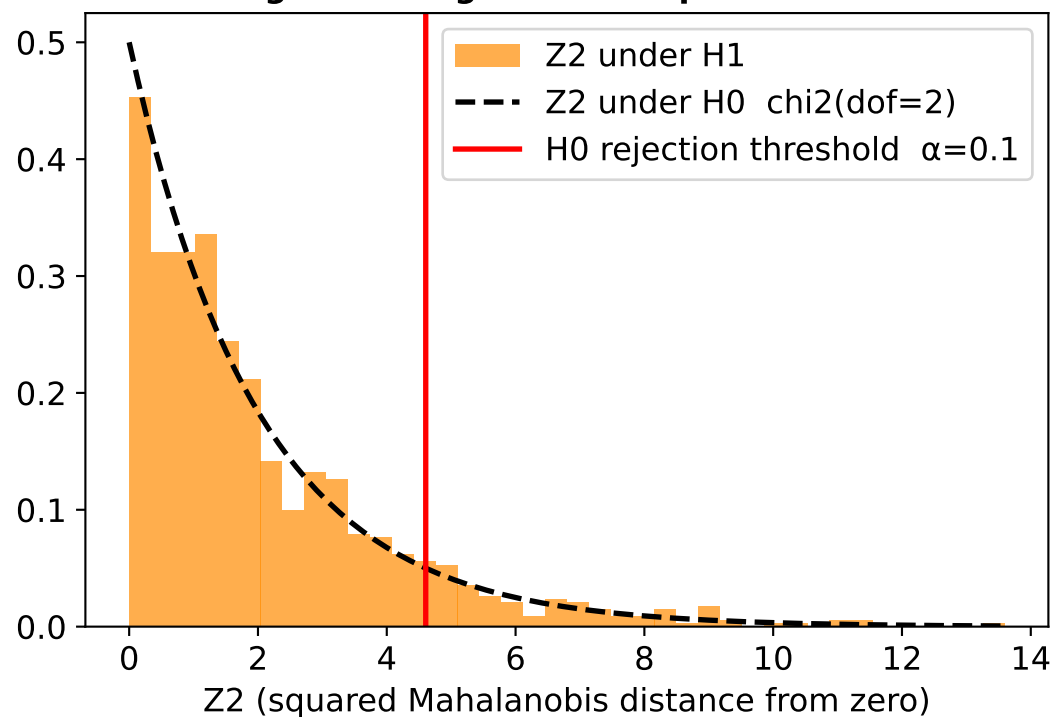
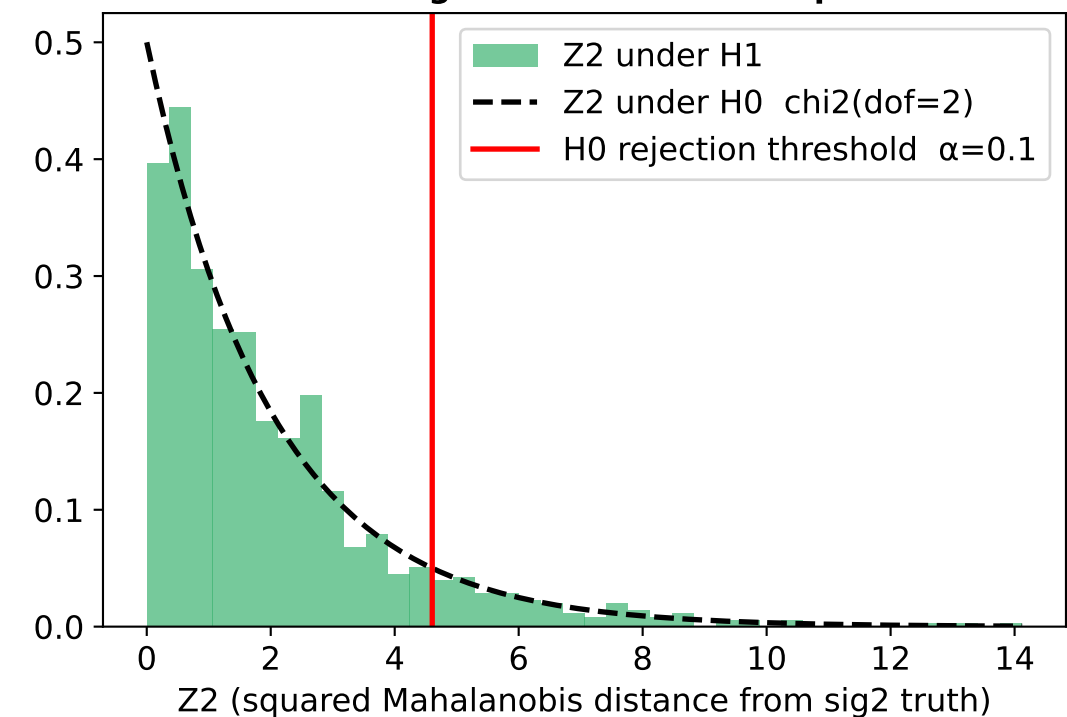
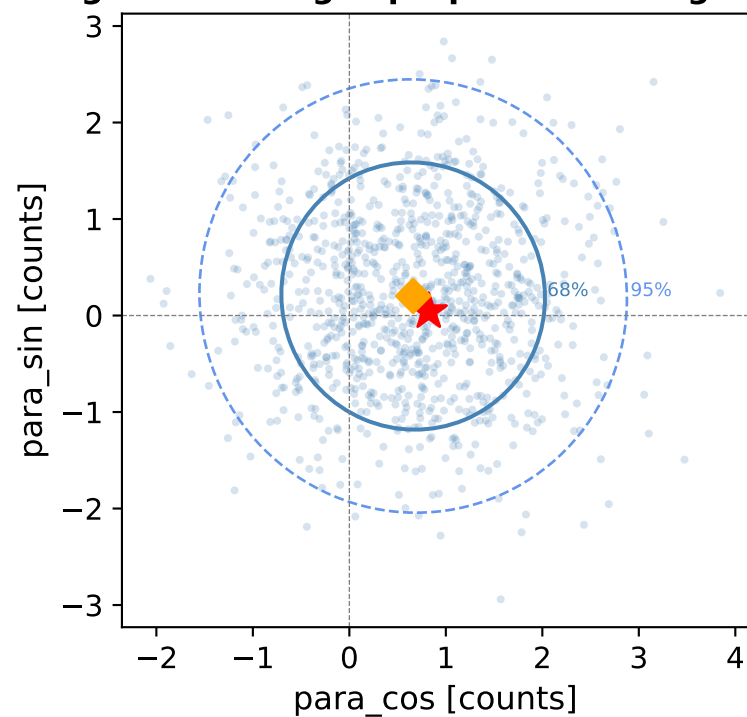
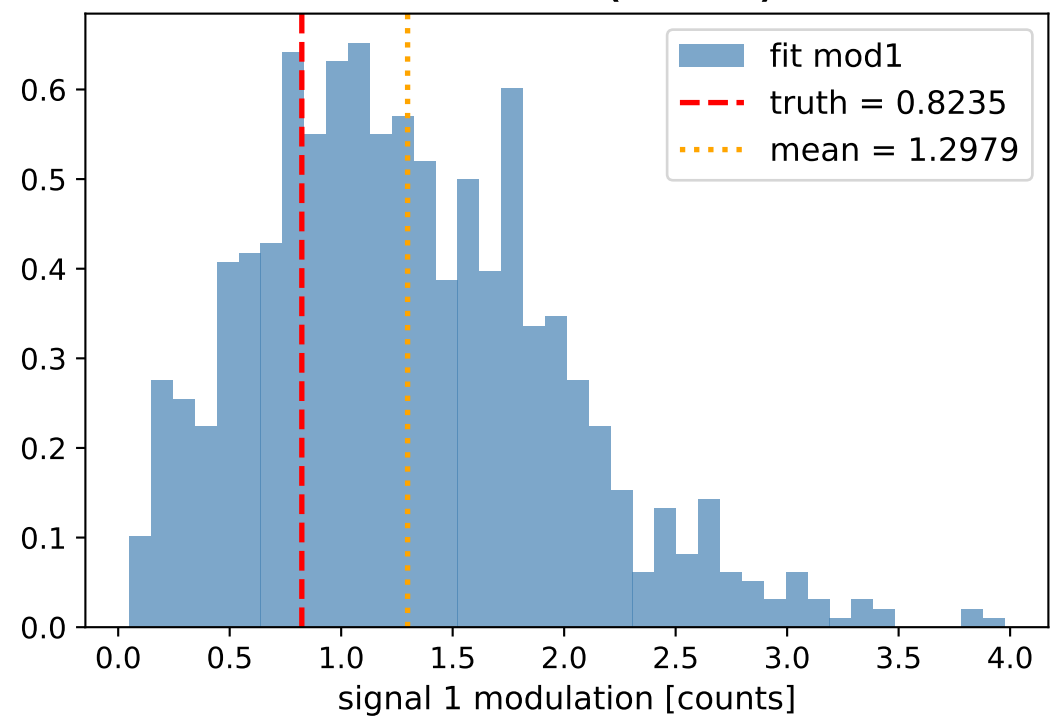
Noiseless input time series

**Run parameters**

$R_{s1} = 30 \text{ t}^{-1}\text{yr}^{-1}$ $R_{s2} = 30 \text{ t}^{-1}\text{yr}^{-1}$
 $R_b = 0 \text{ t}^{-1}\text{yr}^{-1}$
 $D = 10 \text{ t}$ $dt = 30 \text{ day}$ $T = 10 \text{ yr}$
 $\phi_{i1} = 0.0516 \text{ rad}$ $\phi_{i2} = 2.5804 \text{ rad}$
 $N_1 = 24.6407 \text{ counts in bin (unitless)}$
 $N_2 = 24.6407 \text{ counts in bin (unitless)}$
 $N_b = 0.0000 \text{ counts in bin (unitless)}$
 $Ad_1 = 0.03342$ $Ad_2 = 0.01000$
 $N_{\text{toys}} = 1000$ noise = poisson

sig2 fitted modulation**sig2 fitted phase
(uniform → buried; peaked → detectable)**

★ truth ◆ mean fit

sig2 recovered after split SNR=0.19**is signal2 being detected? power = 0.104****is reconstructed signal2 close to truth? power = 0.100****sig1 fit treating superposition as sig1 only****sig1 amplitude bias (ignoring sig2)
 $\Delta = +0.4744$ (+57.6%)****sig1 phase bias (ignoring sig2)
 $\Delta = +8.714^\circ$** 